

Sixth day of class - 9/14

Last time: We talked about projective space, defining lines and curves in projective space, what the points at infinity are, how to reduce a point in projective space mod p , and constructing tangent lines.

Today: We'll talk a bit about intersection multiplicity, singular and non-singular points, and transforming equations of curves.

One more example (if time): Can you use modular arithmetic to prove that there are no rational solutions to

$$x^5 + y^5 = 3?$$

Intersection multiplicity

Definition. The “intersection index” or “multiplicity” of a point P on two curves C_1 and C_2 is a non-negative integer $I(C_1 \cap C_2, P)$ reflecting the extent to which C_1 and C_2 are tangent to one another at P , or one or the other is not smooth at P .

Bezout's Theorem: If C_1 and C_2 are two projective curves, and C_1 and C_2 don't have any components in common, then

$$\sum_{P \in C_1(\mathbb{C}) \cap C_2(\mathbb{C})} I(C_1 \cap C_2, P) = (\deg C_1)(\deg C_2).$$

Example: Let $f(x, y, z) = y^2z - x^3 - z^3$ and $P = (2 : 3 : 1)$.

We have

$$\begin{aligned}\frac{\partial f}{\partial x} &= -3x^2 \\ \frac{\partial f}{\partial y} &= 2yz \\ \frac{\partial f}{\partial z} &= y^2 - 3z^2.\end{aligned}$$

Plugging in $x = 2$, $y = 3$ and $z = 1$ we get

$$\begin{aligned}\frac{\partial f}{\partial x}(0, 1, 1) &= -12 \\ \frac{\partial f}{\partial y}(0, 1, 1) &= 6 \\ \frac{\partial f}{\partial z}(0, 1, 1) &= 9 - 3 = 6.\end{aligned}$$

So the equation of our line is

$$\begin{aligned} -12(x-2) + 6(y-3) + 6(z-1) &= 0 \\ -12x + 24 + 6y - 18 + 6z - 6 &= 0. \end{aligned}$$

The constant term cancels (at it must)!

We can rewrite this as $-2x + y + z = 0$. The intersection of this line with the curve is where $y = 2x - z$. This gives

$$\begin{aligned} (2x - z)^2 z - x^3 - z^3 &= 0 \\ 4x^2 z - 4xz^2 + z^3 - x^3 - z^3 &= 0 \\ -x^3 + 4x^2 z - 4xz^2 &= 0 \\ -x(x^2 - 4xz + 4z^2) &= 0 \\ -x(x - 2z)^2 &= 0. \end{aligned}$$

We see that $x - 2z$ is squared and so this line intersects the curve at the point $x = 2z$, $y = 2x - z = 2(2z) - z = 3z$ at $(2z : 3z : z) = (2 : 3 : 1)$ with multiplicity two.

The other intersection point is $x = 0$. This gives $y = -z$ and so $(x : y : z) = (0 : -z : z) = (0 : -1 : 1)$.

This is saying that if $P = (2 : 3 : 1)$, then $P * P = (0 : -1 : 1)$ and so $P + P = (0 : 1 : 1)$.

2. $x + y + 2 = 0$ and $x^2 + y^2 = 1$.

If we take the first equation and solve for y , we get $y = -2 - x$. Plugging this in gives $x^2 + (-2 - x)^2 = 1$. This gives $2x^2 + 4x + 4 = 1$. This has no real solutions. However, there are two complex solutions. This is what we expect - a line and a conic should intersect in two points.

3. $x + 1 = 0$ and $x^2 - y = 0$.

The line $x = -1$ intersects $x^2 - y = 0$ when $y = (-1)^2 = 1$. Thus, $(-1, 1)$ is a point of intersection. But a conic and a line should intersect in two points. Where's the second point?

(Working projectively, we have $x + z = 0$ and $yz = x^2$. This gives $x = -z$ and so $yz = z^2$. This means that either $z \neq 0$ and $y = z$, $x = -z$ so we get $(-1 : 1 : 1)$, or $z = 0$ in which case $x = 0$ and we get $(0 : 1 : 0)$, a point at infinity.)

These two curves intersect at the point at infinity in the "no slope" direction because both have tangent (or asymptotic) tangent directions.

Singularity

A singular point on a curve C is a point P at which there is no well-defined tangent line. There are several versions of this definition, which are equivalent. We talked about this in the context of genus 0 curves. Here's a projective version.

Definition. If $C : f(x, y, z) = 0$ is a curve in the projective plane, then a singular point is a point $P = (a : b : c)$ where

$$f(a, b, c) = \frac{\partial f}{\partial x}(a, b, c) = \frac{\partial f}{\partial y}(a, b, c) = \frac{\partial f}{\partial z}(a, b, c) = 0.$$

Note: A curve C is called non-singular if there are no singular points in $C(\mathbb{C})$.

Example 1: $y^2 = x^3 + x^2$. We differentiate implicitly and get $2y \frac{dy}{dx} = 3x^2 + 2x$. So

$$\frac{dy}{dx} = \frac{3x^2 + 2x}{2y}.$$

However, if we try to use this definition at the point $(0, 0)$, we get $0/0$ in our fraction. (If we try this at the point $(-1, 0)$, we get $1/0$, which indicates that the tangent line at this point is well-defined and vertical.)

What's going on? (Draw picture)

Example 2: Let $C : f(x, y, z) = y^2z - x^3 + 3xz^2 - 2z^3 = 0$. Are there any singular points on C ?

We have singular points when $f = \frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = \frac{\partial f}{\partial z} = 0$. We get

$$\begin{aligned} \frac{\partial f}{\partial x} &= -3x^2 + 3z^2 \\ \frac{\partial f}{\partial y} &= 2yz \\ \frac{\partial f}{\partial z} &= y^2 + 6xz - 6z^2. \end{aligned}$$

The second equation forces $y = 0$ or $z = 0$. If $z = 0$, then $x^3 = 0$ so $x = 0$ and we get the point $(0 : 1 : 0)$. However, for this point $\frac{\partial f}{\partial z} = 1$. So there are no singular points with $z = 0$.

If $y = 0$, we get $3x^2 = 3z^2$ and $6xz = 6z^2$. We know that $z \neq 0$ and so we can take $z = 1$. This gives $3x^2 = 3$, and $6x = 6$. Thus, $x = 1$. So the point $(1 : 0 : 1)$ is a singular point, and it's the only one.

Definition. An elliptic curve over a field F is a cubic curve C that has at least one point in $C(F)$, and that is non-singular.

(So the curve $E : y^2 = x^3 - 3x + 2$ is not an elliptic curve, because it is singular. The curve $y^2 = f(x)$ is singular precisely when $f(x)$ has multiple roots.)

Warning: If $E/\mathbb{Q}$ is an elliptic curve, it must be non-singular over $\mathbb{Q}$. However, you can take an elliptic curve and reduce it mod p and have it be singular mod p . (This only happens for finitely many p .)

Transforming equations

Any genus 1 curve C over a field F for which there is a non-singular point in $C(F)$ can be transformed into the form

$$y^2 = x^3 + ax + b.$$

This is called short Weierstrass form.

This part of the power of elliptic curves. Here are some examples of types of curves that have genus 1:

- A non-singular curve of the form $y^2 = ax^4 + bx^3 + cx^2 + dx + e$.
- Any non-singular cubic curve. For example,

$$a(a+b)(a+c) + b(a+b)(b+c) + c(a+c)(b+c) = 4(a+b)(a+c)(b+c)$$

is the “fruit equation” I talked about on day 1.

- A non-singular intersection

$$Q_1(x, y, z, w) = Q_2(x, y, z, w) = 0$$

where Q_1 and Q_2 are homogeneous polynomials of degree 2. This curve would be in $\mathbb{P}^3$.

For example,

$$y^2 - 3x^2 = w^2 \text{ and } 2z^2 - y^2 = w^2.$$

The affine points are those of the form $y^2 - 3x^2 = 1$, $2z^2 - y^2 = 1$. This parametrizes rational numbers $n = 3x^2$ for which n is 3 times a square, $n + 1$ is a square, and $n + 2$ is twice a square.

(One such n is 48.)

Maps between curves

We say that a curve C is defined over the field F if its equation (or equations) have coefficients coming from F .

Definition. Suppose that we have two curves C_1 and C_2 are two curves defined over F that live in $\mathbb{P}^2$. A rational map from C_1 to C_2 that is defined over F is a function ϕ so that

- (1) For any point in $P \in C_1(\mathbb{C})$, either $\phi(P)$ is a point in $C_2(\mathbb{C})$, either $\phi(P) \in C_2(\mathbb{C})$, or $\phi(P)$ is undefined, and
- (2) The function ϕ is (locally) given by polynomials with coefficients in F , that is

$$\phi(x : y : z) = (f_1(x, y, z) : f_2(x, y, z) : f_3(x, y, z))$$

where f_1 , f_2 , and f_3 are homogeneous polynomials.

Q1: How could $\phi(P)$ be undefined?

Ex: Let $C_1 : y^2z = x^3 - 2xz^2$ and $C_2 : \mathbb{P}^1$. Define $\phi : C_1 \rightarrow C_2$ by

$$\phi(x : y : z) = (y : x).$$

This is a rational map defined over $\mathbb{Q}$. The point $(0 : 0 : 1)$ is on C_1 , but if we plug this into the formula we get

$$\phi(0 : 0 : 1) = (0 : 0).$$

But $(0 : 0)$ is excluded from $\mathbb{P}^1$.

Q2: What do you mean when you say that $\phi(P)$ is locally given by polynomials?

We can “fix” the last example. If we take

$$y^2z = x^3 - 2xz^2,$$

we can divide both sides by xyz and get

$$\frac{y^2z}{xyz} = \frac{x^3 - 2xz^2}{xyz}$$

and this simplifies to

$$\frac{y}{x} = \frac{x^2 - 2z^2}{yz}.$$

So (as long as $x \neq 0$), $(y : x) = (x^2 - 2z^2 : yz)$.

So we can define

$$\phi_2(x : y : z) = \begin{cases} (y : x) & \text{if } (y : x) \neq (0 : 0) \\ (x^2 - 2z^2 : yz) & \text{if } (x^2 - 2z^2 : yz) \neq (0 : 0). \end{cases}$$

Here we have two different defining equations that we use, and those equations are equivalent (when they are both defined).

We have $\phi_2(0 : 0 : 1) = (0^2 - 2 \cdot 1^2 : 0 \cdot 1) = (-2 : 0) = (1 : 0)$.

The function ϕ_2 is what is called a *morphism*.

Definition. Suppose that we have two curves C_1 and C_2 are two curves defined over F that live in $\mathbb{P}^2$. A morphism from C_1 to C_2 that is defined over F is a function ϕ so that

- (1) For any point in $P \in C_1(\mathbb{C})$, $\phi(P) \in C_2(\mathbb{C})$, or $\phi(P)$ is undefined, and
- (2) The function ϕ is (locally) given by polynomials with coefficients in F .

Theorem. If C_1 and C_2 are curves defined over F and $\phi : C_1 \rightarrow C_2$ is a rational map defined over F , then ϕ can be extended in a unique way to a morphism defined over F .

Definition. Two curves C_1 and C_2 are said to be isomorphic if there is a morphism $\phi : C_1 \rightarrow C_2$ that has an inverse $\phi^{-1} : C_2 \rightarrow C_1$ which is also a morphism.

Ex: Let C_1 be the curve with equations $y^2 - 3x^2 = w^2$ and $2z^2 - y^2 = w^2$, and let $E : y^2 = x^3 - 36x$. Then C_1 is isomorphic to E .

Define $\phi : C_1 \rightarrow E$ by

$$\phi(x : y : z : w) = (3y - 3w : -18x : (-1/2)y + z - (1/2)w).$$

The inverse of ϕ is

$$\phi^{-1}(x : y : z) = (-4yz : x^2 + 12xz - 36z^2 : x^2 + 36z^2 : x^2 - 12xz - 36z^2).$$

(Technically, the formulas I wrote for ϕ and ϕ^{-1} are both rational maps, but these can be extended to morphisms by the previous theorem.)

The conclusion is that the map between the rational points on C_1 and the rational points on E

$$\phi : C_1(\mathbb{Q}) \rightarrow E(\mathbb{Q})$$

is a bijection.

Q: How do we come up with these maps?

Section 1.3

How do we transform a cubic curve C with a rational point P on it into $y^2 = x^3 + Ax + B$?

General strategy:

1. Let $Z = 0$ be the equation of the tangent line to C at P .

The multiplicity of the intersection of the tangent line with C at P is either 2, in which case the tangent line intersects the curve in a second point Q , or it is 3, in which case there is no other such point.

2. If the multiplicity is 2, let $X = 0$ be the tangent line through Q . If the multiplicity is 3, let $X = 0$ be any line that doesn't have P on it.
3. Let $Y = 0$ be any line (other than $Z = 0$) that goes through P .
4. Setting $x_1 = X/Z$ and $y_1 = Y/Z$ we get an equation of the form

$$x_1 y_1^2 + (ax + b)y_1 = cx_1^2 + dx_1 + e.$$

We multiply through by x and get

$$x_1^2 y_1^2 + (ax_1 + b)x_1 y_1 = cx_1^3 + dx_1^2 + ex_1.$$

We set $y_2 = x_1 y_1$ and get

$$y_2^2 + (ax_1 + b)y_2 = cx_1^3 + dx_1^2 + ex_1.$$

We then complete the square in y_2 to get

$$y_2^2 + (ax_1 + b)y_2 + \frac{(ax_1 + b)^2}{4} = cx_1^3 + dx_1^2 + ex_1 + \frac{(ax_1 + b)^2}{4}$$

and we set $y_3 = y_2 + \frac{ax_1 + b}{2}$. We now have $y_3^2 = cx_1^3 + fx_1^2 + gx_1 + h$.

If we multiply through by c^2 , we get $c^2 y_3^2 = c^3 x_1^3 + f x_1^2 + g x_1 + h$. We set $y = c y_3$ and $x_2 = c x_1$ we get

$$y^2 = x_2^3 + \frac{f}{c^2} x_2^2 + \frac{g}{c} x_2 + h.$$

Now, the cubic $x_2^3 + (f/c^2)x_2^2 + \frac{g}{c}x_2 + h$ has three roots (over $\mathbb{C}$), and the sum of the roots is $-f/c^2$.

By setting $x = x_2 - \frac{f}{3c^2}$, the roots (in x) are now larger by $f/3c^2$, and so the sum of the roots of the polynomial in x is zero. This gives us an equation of the form

$$y^2 = x^3 + Ax + B.$$